The Symmetric Group

<u>Definition (Bijections).</u>

For two sets X and Y, a map $f: X \to Y$ is a bijection if it is both

(1) Injective (one-to-one): different x values to different y values.

(2) Surjective (onto): every element of Y is in the image of f.

<u>Definition (Permutation).</u> A permutation of a set X is a bijection $f: X \to X$.

<u>Example.</u> $X = \{a, b, c, d\}$ and $f: X \to X$ defined by

$$f: \begin{array}{ll} a \mapsto b & c \mapsto a \\ b \mapsto c & d \mapsto d \end{array}$$

<u>Definition.</u> Given a permutation $f: X \to X$, the inverse permutation $f^{-1}: X \to X$ is the inverse map defined by $f^{-1}(w) = x$ if $f(x) = w$.

<u>Example.</u> For f as above

$$f^{-1}: \begin{array}{ll} a \mapsto c & c \mapsto b \\ b \mapsto a & d \mapsto d \end{array} \Bigg\} \text{ Inverses only make sense for bijections}$$

Key Properties of permutations

① If f, g are permutations, then $f \circ g$ is also a permutation.

② Composition of maps is associative.

③ The identity map $i_X : X \to X$, $x \mapsto x$ is a permutation.

④ Composing f, f^{-1} gives the identity map i_X.

Conclusion: The set of permutations on a set X form a group with operation given by composition.

Definition (Symmetric Groups).

The symmetric group on n elements, denoted by S_n or $\mathrm{Sym}(n)$, is group of permutations

on the set $\{1, 2, 3, \ldots, n\}$.

Fact: $|S_n| = \#\{\text{permutations in } S_n\} = n!$

Proof. There are n choices for where 1 maps to

$$\begin{array}{cc} n-1 & 2 \\ n-2 & 3 \\ \vdots & \vdots \end{array}$$

Total: $n(n-1)(n-2)\cdots 2\cdot 1 = n!$

<u>Cycle Notation</u>

<u>Example.</u> $\begin{array}{|cccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \hline 3 & 5 & 2 & 6 & 1 & 8 & 7 & 4 \end{array}$, in S_8

$1 \to 3 \quad 2 \to 5 \quad 3 \to 2 \quad 4 \to 6 \quad 5 \to 1 \quad 6 \to 8 \quad 7 \to 7 \quad 8 \to 4$

$1 \to 3 \to 2 \to 5 \qquad 4 \to 6 \to 8 \qquad 7$

We call these cycles and write them as

$$(1\ 3\ 2\ 5)(4\ 6\ 8)(7).$$

Any permutation can be written as disjoint cycles.

Computing Products

<u>Example</u>. In S_3, if $\alpha = (1\ 3)$, $\beta = (1\ 2\ 3)$ then the product $\alpha\beta$ is

$$\left.\begin{array}{l} 1 \to 3 \to 1 \\ 2 \to 2 \to 3 \\ 3 \to 1 \to 2 \\ \quad\ \alpha \quad \beta \end{array}\right\} \Rightarrow \begin{array}{l} 1 \to 1 \\ 2 \to 3 \\ 3 \to 2 \\ \quad \alpha\beta \end{array} \qquad \alpha\beta = (2\ 3)$$

Transpositions

<u>Definition (Transpositions)</u>.

A transposition $\tau \in S_n$ is a permutation which only swaps two elements, i.e.

$$\tau = (a\ b), \quad \text{for some } a, b.$$

We call it a simple transpositions if it has a cycle notation

$$\tau = (i\ \ i+1)$$

<u>Proposition</u>. Any cycle can be written as a product of transpositions.

Proof. $(a_1\ a_2 \cdots a_k) = (a_1\ a_2)(a_1\ a_3)\cdots(a_1\ a_k)$.

<u>Corollary</u>. As a permutation is a product of disjoint cycles, any permutation is a product of transpositions.

<u>Example</u>. $\alpha = (1\ 3\ 5)(2\ 4\ 6)$
$= (1\ 3)(1\ 5)(2\ 4)(2\ 6)$

<u>Definition (Parity)</u>.

We say a permutation α is even / odd if it can be written as a product of an even / odd number of transpositions.

<u>Theorem</u>. Every permutation α in S_n is either even or odd and not both.

Terminology: We call this the parity of α.

Remark.

1) A cycle $(a_1\ a_2 \cdots a_k)$ has parity

$$\begin{cases} \text{even}, & k \text{ is odd} \\ \text{odd}, & k \text{ is even}. \end{cases}$$

2) Transpositions are self-inverse: $\tau = (a\ b)$,

$$\tau^{-1} = (a\ b) \Rightarrow \tau = \tau^{-1}.$$

3) A permutation α and its inverse α^{-1} have same parity.

Proof. 3) $\alpha = \tau_1 \tau_2 \cdots \tau_k$

$$\Rightarrow \alpha^{-1} = \tau_k^{-1} \tau_{k-1}^{-1} \cdots \tau_2^{-1} \tau_1^{-1}$$
$$= \tau_k \tau_{k-1} \cdots \tau_2 \tau_1$$

Subgroups

Definition (Subgroups).

Let $(G, *)$ be a group. Let $H \subseteq G$ be a subset. Then we say H is a subgroup of G if $(H, *)$ is a group with the same operation. We denote this by $H \leq G$.

Example. 1) $G \leq G$.

2) $\{e\} \leq G$.

3) $H = \{0, 2, 4, 6\} \subseteq \mathbb{Z}_8$, then $(H, +) \leq (\mathbb{Z}_8, +)$

4) $(\mathbb{R} \setminus \{0\}, \cdot) \not\leq (\mathbb{R}, +)$ (different operations).

A key step to show that $H \leq G$ is to show that H is "closed under $*$", i.e. $\forall a, b \in H$, $a * b \in H$.

Useful shortcut: If H is finite, then to prove $H \leq G$, we only need to show

- H is non-empty
- H is closed under $*$.

Definition (The Alternating Group).

The set $Alt(n)$ or A_n is defined by

$$\{\alpha \in S_n \mid \alpha \text{ even}\}.$$

We call this the alternating group on n elements.

Proposition. $A_n \leq S_n$.

Proof.

1) Nonempty: $e \in S_n$ is even, so $e \in A_n$.
2) Closed: For even permutations α, β,

$$\alpha = \tau_1 \tau_2 \cdots \tau_{2n}, \quad \beta = \eta_1 \eta_2 \cdots \eta_{2k}$$

$$\Rightarrow \alpha\beta = \underbrace{\tau_1 \cdots \tau_{2n} \eta_1 \cdots \eta_{2k}}_{2(n+k)}$$

Fact: $|A_n| = \frac{1}{2}|S_n| = \frac{1}{2}(n!)$